

2(a)	<i>any two from:</i> 1 no melanin ; 2 white hair ; 3 (very) pale skin ; 4 pink / very pale / AW, eyes ;	2
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Question	Answer	Marks
2(b)(i)	any three from: 1 more males / fewer females, have ocular albinism ; 2 males with (one copy of) recessive allele have ocular albinism / females with (one copy of) recessive allele are carriers ; 3 males (with ocular albinism) receive recessive allele from mothers ; 4 fathers cannot pass on ocular albinism to their sons ; 5 (because) males pass Y chromosome to sons ; 6 males with ocular albinism pass recessive allele to daughters ; 7 detail from Fig. 2.1 ; e.g. all daughters (3, 5, 7) of male 1 are carriers / all sons (9 and 10) of male 1 do not have ocular albinism / males 11, 12, 13, get ocular albinism from mothers 3 and 5	3
2(b)(ii)	symbols normal <u>allele</u> = A OA1 <u>allele</u> = a ; parental genotypes X^aY x X^AX^A and gametes X^a Y and (all) X^A ; offspring genotypes (phenotypes correctly linked) X^AX^a X^AY ; normal / carrier, female normal male ;	4

Question	Answer	Marks
2(b)(iii)	any two from: 1 frameshift / every triplet from site of deletion is changed ; 2 primary structure / tertiary structure / 3D shape, is changed or (protein) folded incorrectly; 3 (protein has) changed, binding / active, site; 4 premature STOP codon leads to incomplete protein ;	2
2(b)(iv)	any one from: for 1 decide whether to have children ; 2 earlier treatment if child born with ocular albinism ; any one from: against 3 test may be expensive ; 4 condition, is mild / not life threatening ; 5 can't be treated ;	2

Question	Answer	Marks
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